

Computational detection of a high-risk DLBCL group

A summary of my master thesis

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October 15th, 2024

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The goals of my thesis

Develop a binary classifier that predicts the risk of DLBCL patients: high (PFS < 2 years) or low (PFS $\geq$ 2 years).

In more detail:

- **Performance:** rate of positive predictions (or prevalence) ≥ 0.10 and precision significantly (95%-CI) above the precision of the IPI on an independent, prospective data set. Make the model transferable from one data set to another.
- **Heuristics and recommendations for the future:** Which model classes for which hyperparameters are worth training and validating? Which models can we validate reliably? Which perform strongly on new data? What are the requirements for the training and test data?

A quick dive into the methods

The big picture

I did the well-known train-validate-test approach.

- **Training and validation** on the training data. Fit several candidate models, validate them (cross validation, out-of-bag predictions) and choose a single model, the best one according to validation.
- **Testing** on the test data. Evaluate the performance of the chosen model.

The smaller picture: candidate models

- All fit models are Gauss, logistic, Cox models or random forests or compositions thereof (late integration of IPI, LAMIS, other already-existent and own signatures).
- For training, I tuned the time cutoff that separates high- and low-risk patients.
- A-priori feature selection: Include gene-expression levels? Format of certain features?
- Loss-function-based feature selection: elastic-net, usually LASSO regularization.
- Enforce zero-sum constraint for gene-expression levels to gain cross-platform transferability [1; 2].
- Add combinations of at most n_{combi} categorical features to the predictor if sufficiently prevalent.

The results

Meet the data

data set	Schmitz [4]	Reddy [3]	Staiger [5]
prospective trial	no	no	yes
response	PFS < 2	OS < 2.5	PFS < 2
# samples	229	604	466
# genes with expression level	25 066	13 302	145
technology	RNA-seq	RNA-seq	NanoString
high risk [%]	36.6	31.5	24.3
prev(tlPI) [%]	12.9	21.6	17.0
prec(tlPI) [%]	65.2	54.1	38.2

Table 1: All data sets include the five IPI features in their thresholded format, the IPI score and group, gender, cell of origin and the LAMIS.

For every of the three data sets: use it as training data, the respective remaining two data sets as test data.

	Schmitz	Reddy	Staiger
Schmitz	12.9, 65.2	17.7, 59.6	17.1, 50.7
Reddy	17.8, 71.1	21.6, 54.1	18.0, 53.2
Staiger	17.4, 75.7	22.5, 50.4	17.0, 38.2

Table 2: Prevalence and precision. Diagonal entries (i, i) hold prevalence, precision of the tIPI on cohort i . Off-diagonal entries (i, j) hold prevalence, precision on cohort j of the best model trained and validated on cohort i .

Dito for lower limit of 95%-CI of precision

	Schmitz	Reddy	Staiger
Schmitz	65.2	48.6	38.7
Reddy	54.1	54.1	41.5
Staiger	58.5	40.9	38.2

Table 3: Lower limit of the 95%-confidence interval of the precision. Diagonal entries (i, i) hold the precision of the tIPI on cohort i . Off-diagonal entries hold the lower limit of the 95%-confidence interval of the precision on cohort j of the best model trained and validated on cohort i .

In the following, we focus on the best model trained and validated on the Reddy cohort, denoted m_{Reddy}^* .

What does m_{Reddy}^* look like?

A logistic model, trained with LASSO regularization, the time cutoff separating into low- and high-risk patients is 2.3 (versus 2.5).

Features:

- No gene-expression levels a-priori directly included.
- Single continuous feature included a priori: LAMIS score.
- Categorical features: gender, LDH ratio > 1, number of extranodal sites > 1, Ann Arbor stage > 1, age > 60, ECOG > 1, IPI group, and combinations of two of them.
- Sparse model: 13 features have non-zero coefficients.

How reliable is the validation for the models trained on the Reddy data?

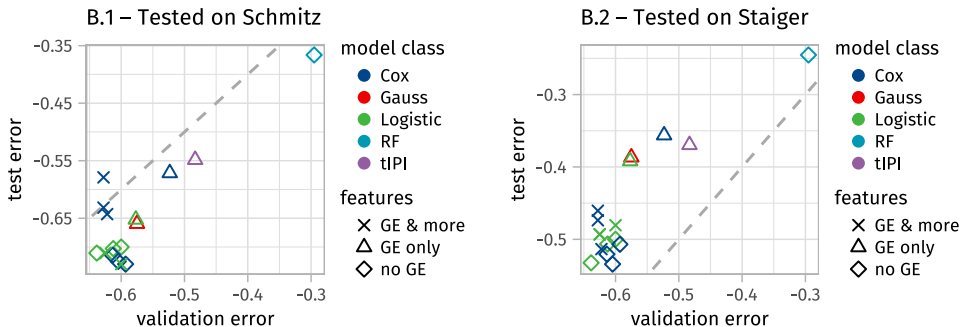


Figure 1: Validation versus test error. The dashed gray line is the identity line. The training survival cutoff T , n_{combi} , elastic-net regularization strength λ (for GLMs), feature subsample size m and minimal node size n_{min} (for RFs) have already been optimized with respect to the validation error.

On thresholding the continuous output of m_{Reddy}^*

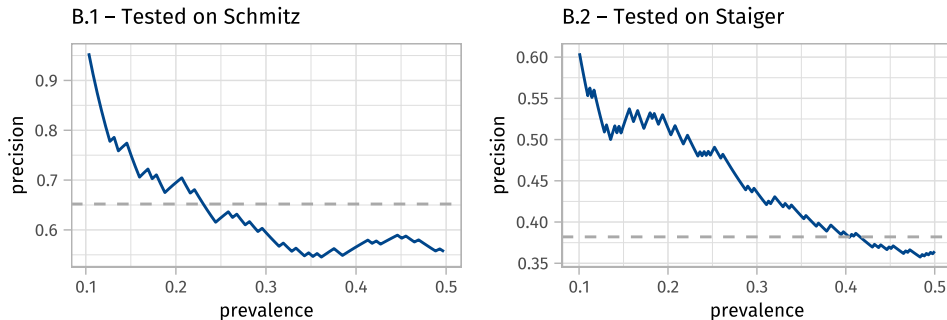


Figure 2: Prevalence versus precision for all possible ways to threshold m_{Reddy}^* for a prevalence in $[0.10, 0.50]$. The dashed gray line marks the precision of the tIPI on the respective cohort.

Dito for the lower limit of the 95%-CI of the precision

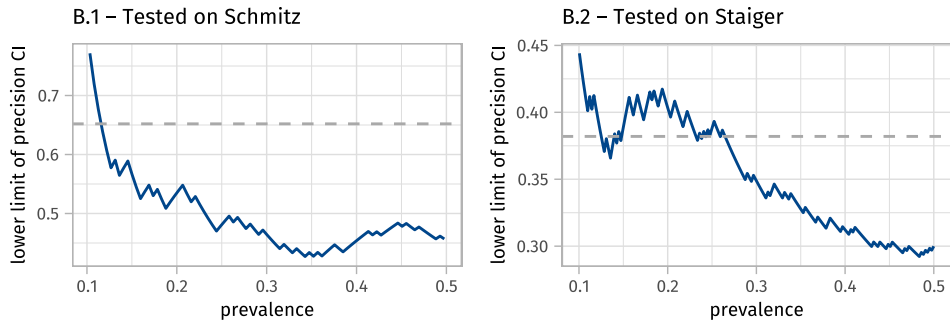


Figure 3: Prevalence versus lower limit of the 95%-CI of the precision for all possible ways to threshold m_{Reddy}^* for a prevalence in $[0.10, 0.50]$. The dashed gray line marks the precision of the tIPI on the respective cohort.

Find the details on <https://github.com/lgessler/master-thesis>.

Thank you! Questions?

References

- [1] M. Altenbuchinger, T. Rehberg, H. U. Zacharias, F. Stämmeler, et al. Reference point insensitive molecular data analysis. *Bioinformatics*, 33(2):219–226, 09 2016. ISSN 1367-4803. doi: 10.1093/bioinformatics/btw598.
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- [3] A. Reddy, J. Zhang, N. S. Davis, A. B. Moffitt, et al. Genetic and functional drivers of diffuse large B cell lymphoma. *Cell*, 171(2):481–494.e15, 2017. ISSN 0092-8674. doi: 10.1016/j.cell.2017.09.027.
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